Project Expo 3-11-26

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Department of Computer Science & Engineering
(IoT, Cyber Security including Blockchain Technology)

NAAC A+ | NBA CSE ECE ISE

Title of the Project – Design and Development of Camera Based Smart Parking System with Real-Time Slot Monitoring and Dashboard Visualization

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Batch No. – Batch-04



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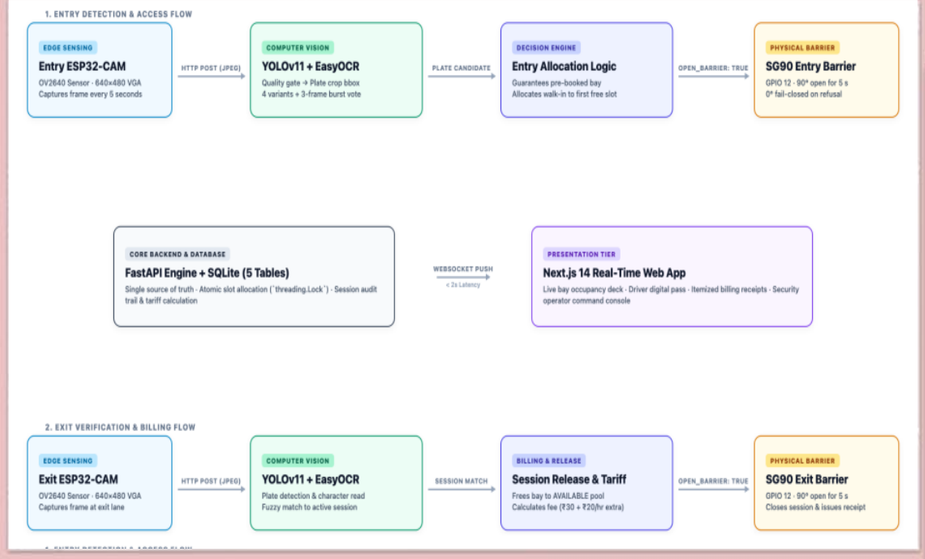


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METHODOLOGY & WORK FLOW DIAGRAMS



WORK COMPLETED

1. Built two ESP32-CAM gate nodes for entry and exit.
2. Integrated YOLOv11 + EasyOCR with four variants and three-frame voting.
3. Implemented six-bay booking, walk-in entry and automatic bay allocation.
4. Added two SG90 fail-closed barriers and session-based billing.
5. Built FastAPI + SQLite backend and Next.js 14 dashboard, passes and receipts.
6. Passed 15/15 automated checks, 8/8 UI checks and HIL flows.

AIM OF THE PROJECT

To design and implement a camera-based smart parking system that automates vehicle access, bay allocation, and real-time monitoring using dual-gate, eliminating the need for per-bay sensors.

INTRODUCTION

1. Drivers waste significant time and fuel searching for parking due to a complete lack of real-time bay visibility.
2. Traditional smart parking installs sensors in every bay, causing hardware, wiring, and maintenance costs to explode as lots grow.
3. AutoPark shifts sensing entirely to the gates—monitoring the entire facility using just two camera nodes with zero per-bay sensors.
4. An edge vision pipeline (YOLOv11 + EasyOCR) reads plates, verifies reservations, and triggers automated boom barriers only for authorized cars.
5. A centralized backend tracks live occupancy and instantly updates driver passes, billing receipts, and operator dashboards via WebSockets.

OBJECTIVES

Objective 1 : Detect and recognize vehicle license plates at gate checkpoints using YOLOv11 and Easy OCR.

Objective 2 : Guarantee pre-booked parking bays and automatically assign available slots to walk-in vehicles.

Objective 3 : Prevent unauthorized access by rejecting low-confidence plates and controlling fail-closed servo boom barriers.

Objective 4 : Infer lot occupancy from gate events and provide real-time dashboards, digital passes, and automated receipts.

Objective 5 : Validate system reliability and decision logic through automated test suites, UI workflows, and hardware simulation.

CONCLUSIONS

- End-to-end ANPR, booking, walk-in allocation, entry/exit, barrier actuation and billing workflow demonstrated.
- Two gate cameras replace per-bay sensors; sensing cost remains constant as bays increase..
- Occupancy depends on recognition quality; manual release and audit trail handle exceptions.
- Paper presented at Scopus-indexed conference.

DESIGN & SIMULATION TOOLS

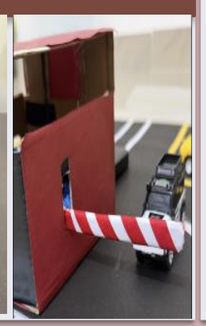
1. ESP32-CAM / Arduino IDE
2. Python 3.13 + FastAPI
3. YOLOv11 + EasyOCR + OpenCV
4. SQLite + WebSocket
5. Next.js 14, JWT, bcrypt, pytest

PROJECT PHOTOGRAPHS / SNAPSHOTS









OUTCOME OF THE PROJECT

1. Working six-bay prototype with ANPR entry and exit.
2. Automatic bay allocation for booked and walk-in vehicles.
3. Fail-closed barrier control with refusal on unclear reads.
4. Live dashboard, passes, itemised receipts and operator console.
5. Two gate cameras replace per-bay sensors, costs constant regardless of lot capacity.